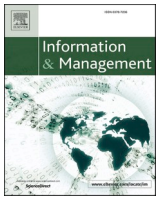
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Unveiling the paradox of NFT digital scarcity: Impacts of digital scarcity on online trading and social behaviors

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ABSTRACT

Embracing the scarcity principle, we explore the effect of digital scarcity in nonfungible tokens (NFTs) on users' online trading behaviors and social interactions (e.g., considering them as gifts). Using a dataset obtained from the NFTGo platform, we find that digital scarcity facilitates social behaviors but hinders trading behaviors. Notably, the influences of digital scarcity are more pronounced when there is less uncertainty (e.g., higher consistency of rarity evaluation and stronger impact of influencers). This research not only enriches the NFT literature with fresh insights into social and trading behaviors, but also extends the scarcity principle to the digital realm.

1. Introduction

Nonfungible tokens (NFTs) are blockchain-based digital assets that authenticate unique ownership of content [1,2]. By leveraging their nonfungibility, NFTs ensure the distinctiveness of each token. By providing verifiable authenticity, ownership, and provenance, NFTs have notably impacted the evolution of Information Systems (IS), particularly in digital asset management [3,4]. However, the broader digital asset market has undergone significant shifts since its 2021 peak. According to the report, the NFT trading volume has collapsed by more than 93 % from its peak of 16.5 billion in Q1 2022 to under 1.1 billion by Q3 2024 [5]. This contraction mirrors systemic declines across cryptocurrency-based markets; for example, the trading volume of Meme coins has plummeted by 82 % from its peak in December 2024 to March 2025 [6]. Simultaneously, AI tokens also experienced a 72 % decline during the same period [7], suggesting structural challenges transcending individual asset classes.

The market's sharp decline stems from converging factors, such as speculative bubble bursts, crypto scams, and technological bottlenecks [8]. These macroeconomic forces have fundamentally reshaped user engagement patterns. At the trading level, global NFT trading volume declined by 76 % in 2024 versus 2022 [9], with high-profile collections like Bored Ape Yacht Club (BAYC) and Azuki depreciating from millions to tens of thousands within months [8]. More critically, more than 79 % of listed NFTs remained unsold in 2024 exchanges [10], indicating severe liquidity erosion. Concurrently, social communities stagnated. The data from NonFungible (NonFungible.com) shows that the number of active users has been continually falling over the past 3 years and hit a new low in 2024 [11], signaling waning speculative interest but potential paradigm shifts in value perception.

As these structural contractions redefine market parameters, digital scarcity gains pivotal status in NFT ecosystems by dually governing value determination and market dynamics. Distinct from traditional time-based and quantity-based scarcity, digital scarcity represents a creator-controlled, attribute-driven form of rarity. This novel scarcity paradigm has a substantial impact on transactional behaviors. It demonstrates measurable effects not only on NFT pricing [12,13] and liquidity [14], but also on market innovation trajectories [15]. Beyond economic implications, digital scarcity actively shapes social behaviors within digital cultural networks by reconstituting user perceptions of value, status differentiation, and community affiliation [16,17]. As a flexible and quantifiable measure of value [13,18], digital scarcity presents a promising framework for facilitating market activity [19–21].

This study examines the dual impacts of digital scarcity on trading behaviors (e.g., buying/selling) and social interactions (e.g., gifting). The digitally native nature of such scarcity creates inherent unpredictability that amplifies decision making uncertainty among users. Given this context, it becomes crucial to identify key determinants that

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enhance user confidence and desirability toward NFT acquisitions. Through systematic analysis of the influence between digital scarcity and behavioral patterns, this research seeks to advance theoretical understanding of digital scarcity's market effects. Concurrently, we investigate contextual moderators (*e.g., consistency of rarity evaluations and impact of influencers*) influencing these relationships, aiming to provide actionable insights for fostering sustainable value creation and guiding the strategic evolution of NFT marketplaces.

While trading behaviors remain the predominant manifestation of NFTs as an asset class, their social utility—particularly through practices like gifting—has emerged as a critical dimension of their cultural significance [22]. This shift is most evident among younger generations: Millennials and Generation Z increasingly leverage NFTs as tools for constructing symbolic capital, transforming gift-giving into curated rituals that redefine digital sociality [23]. These practices transcend mere transactionality, functioning as affiliative acts that reinforce interpersonal bonds and cultivate communal identity [24]. Empirical evidence from a 2024 survey of 1,010 US participants reveals that 32.7 % have engaged in NFT gifting, underscoring its growing cultural salience as a performative act of digital kinship. Beyond reinforcing social ties, this behavioral shift demonstrates a latent potential to stimulate market vitality through enhanced user engagement and network-driven participant acquisition [25,26]. Nevertheless, scholarly attention persists in privileging economic dimensions—exemplified by a disproportionate focus on price [13], return on investment [22], and suspicious trading [27]—while systematically marginalizing the sociocultural innovations reshaping NFT ecosystems.

In the NFT ecosystem, creators define NFT rarity by assigning distinct traits varying probabilities [28]. While researchers have begun exploring the economic implications of digital scarcity, particularly its role in price formation and transaction patterns [12,29], scholarly attention remains narrowly focused on transactional aspects. Empirical evidence shows that increased digital scarcity correlates with decreased trade frequency, thereby constraining market liquidity [13,14]. Given the rising use of NFTs as social gifts, it is critical to investigate how digital scarcity shapes users' perceptions of uniqueness, social identity, and status differentiation, as well as its interplay with social behaviors. Crucially, deepening our understanding of how digital scarcity shapes both transactional and social behaviors is essential. This understanding is key to mitigating market challenges and providing a foundation for effective strategy development [30].

Adopting the scarcity principle as a theoretical foundation, this study examines the dual impact of digital scarcity on users' trading and social behaviors, offering both theoretical insights and practical implications [31]. We find that although digital scarcity fosters social engagement, it can inhibit transaction frequency, potentially hindering market activity. Specifically, our analysis reveals that the stronger impact of influencers mitigates the negative correlation between digital scarcity and trading behavior, while simultaneously amplifying the positive association between digital scarcity and social behavior. Moreover, a higher consistency of rarity evaluation strengthens the dual impact of digital scarcity on both behavioral dimensions [22,32].

This study makes several significant contributions. First, we develop a dual-process model showing how blockchain-based trait scarcity differently affects trading behaviors and social behaviors—a distinction rarely examined in NFT studies. By identifying two key moderating factors, this model also extends NFT research frameworks. Second, we extend the concept of scarcity in blockchain systems by introducing attribute-based digital scarcity [33]. Unlike prior research focused on time-based or quantity-based scarcity [34–36], our framework demonstrates how attribute-based scarcity influences NFT market transactions and social behaviors. Furthermore, by incorporating the digitally-native investment logics inherent to NFTs, we expand the applicability of the scarcity principle in digital environments [37]. Third, our findings not only offer valuable implications for NFT creators and platform managers, but also provide cross-asset scalability frameworks applicable to other tokens (e.g., Meme coins, AI coins) and establish systemic governance paradigms for crypto markets.

2. Related literature

2.1. Literature on NFTs

NFTs are a new category of digital assets that leverage blockchain technology for unique identification and authentication [38,39]. This technology ensures that each NFT is unique and scarce, attracting a growing number of participants to the market [40]. The research focus has shifted from defining NFTs and analyzing market trends to exploring the motivations behind NFT purchases [41]. Understanding these motivations is crucial for analyzing market development dynamics. Studies have highlighted two primary motivations for purchasing NFTs. First, akin to other speculative assets, NFTs are increasingly being seen as financial instruments that promise high investment returns [22,41]. Second, owning NFTs provides social recognition and fulfills self-esteem needs, offering emotional returns [42,43]. These motivations give rise to two distinct user behaviors in the NFT market: trading and social behaviors.

Trading behavior involves the buying and selling of NFTs, driven by their economic value. Beyond their economic value, NFTs hold substantial social and cultural significance [44]. As NFTs diversify and the metaverse expands, they have emerged as key tools for enabling digital social interactions [25]. Users utilize NFTs to strengthen relationships through gifting and as symbols of identity and status, enhancing social recognition [45]. Concurrently, many NFT creators are focusing on community building through innovative NFT designs [46]. Social platforms are increasingly leveraging NFTs to foster user connections and open new pathways for social engagement [47]. Thus, although studying trading behaviors is vital for understanding market dynamics, exploring social behaviors in the NFT context is equally important.

Current research often emphasizes NFT pricing, neglecting in-depth studies of user behaviors [29,48,49]. Although price is a clear indicator of an NFT's value, a deeper understanding of these behaviors is essential for implementing effective platform management strategies. Exploring these behaviors enables a comprehensive understanding of the decision making processes in the NFT market. This insight can guide the design of market mechanisms to better align with user demands and market principles. Additionally, insights into user behaviors enable more accurate predictions of NFT market performance, which aids risk and return assessments. Such understanding also aids NFT creators and operators in identifying user needs and market trends. This in turn fosters improved creative and marketing strategies that enhance user engagement and loyalty, thereby driving the platform's long-term growth.

2.2. Digital scarcity of NFTs

In the NFT community, scarcity serves as a fundamental factor in the determination of value [29]. Unlike traditional scarcity, which is tied to time or quantity, NFT rarity stems from unique attributes. By assigning varying probabilities to different traits, creators can actively control and shape the scarcity of their NFTs. This form of scarcity, derived from the uniqueness of digital properties, is known as "digital scarcity" [4]. Although some prior research has examined the role of digital scarcity in NFTs (see Table 1), most studies have focused on price; for instance, a study of NBA Top Shot by Lee [48] showed that increased rarity leads to higher NFT prices. Specifically, a 1 % reduction in circulation corresponds to an average price increase of 0.74 %. This finding aligns with other research emphasizing the importance of rarity in pricing strategies [22,29]. However, NFTs are not just financial assets tied to price; user behavior plays a crucial role in shaping the market. Especially in the face of declining transaction volumes and stagnating user growth, it is crucial to explore how digital scarcity influences user behavior. Such an

Table 1
Key literature on the digital scarcity of NFTs.

Author	Moderator variable	Dependent variable	User behavior	Social behavior	Conclusion
Lommers and Baioumy, 2023	——	Price	×	×	The rarity of NFTs is the core factor determining their pricing.
Bianchi and Dobrynskaya, 2023	——	Price, return on investment	×	×	Rarity does relate positively with the price, on average across collectibles; The average returns of NFT collectibles are positively related to the rarity scores.
Yuan et al., 2025	Rank, size of the collection, temporality	Trading volume	✓	×	Rarity awareness negatively impacts trading volume.
Mekacher et al., 2022	——	Market performance, return on investment	✓	×	Rarity positively relates to NFT trade prices and inversely relates with trade frequency. Additionally, rarity is linked to higher investment returns and a lower probability of negative returns.
Kong and Lin, 2022	——	Price, investment performance	×	×	The pricing of NFT also largely depends on a token's scarceness and investors' aesthetic preferences.
Lee, 2022	Popularity	Price	×	×	The rarer NFT is, the higher its price tends to be; more popular products tend to be less price-sensitive than less popular ones.
Suchow and Ashrafimoghari,2022	——	Price	×	×	The rarity will become devalued over time if the demand is misunderstood because of sampling bias.
This research	Consistence of rarity evaluation, impact of influencers	Trading behaviors, social behaviors	✓	✓	The digital scarcity of NFTs has a negative impact on users' trading behaviors, while positive impact on social behaviors. In addition, consistency of rarity evaluation and impact of influencers are important moderators.

endeavor can enable creators and platform managers to gain deeper insights into user needs and market dynamics, thereby refining creative content and operating strategies. Furthermore, it has the potential to foster user engagement and loyalty, thereby contributing to the platform's long-term sustainability and growth. Despite its significance, the relationship between digital scarcity and user behavior remains underexplored, presenting a critical shortfall in the literature.

Two significant studies have investigated the relationship between NFT digital scarcity and user behaviors. Yuan et al. [14] demonstrated that introducing a rarity rank causes an 11 % decline in transaction volume. Similarly, Mekacher et al. [13] reported that higher NFT rarity scores correlated with fewer transactions, suggesting a negative impact on trading behaviors. Although these studies highlight the potential of digital scarcity to reduce market liquidity, they fail to consider the boundary conditions that could mitigate this effect. Furthermore, the underlying mechanisms driving these behaviors remain unexplored. These deficiencies underscore the necessity for a more comprehensive investigation into the intricate relationship between scarcity and user activity.

As discussed in Section 2.1, digital scarcity in the NFT market not only establishes value, but also caters to users' desires for uniqueness, identity, and social status. Beyond trading behaviors, NFTs are increasingly being used for social interactions, such as gifting; for instance, platforms like Telegram have incorporated NFT gifting features as a means to boost user engagement. This underscores the need to examine how scarcity shapes social behaviors centered on uniqueness and personal expression. However, as outlined in Table 1, existing NFT research has largely neglected this dimension, leaving a critical gap in understanding the broader behavioral implications of digital scarcity.

2.3. Scarcity principle

The scarcity principle explains how individuals behave when faced with limited resources [31,50]. Scarce items are often seen as more valuable, increasing their desirability and driving acquisition. This principle highlights the fundamental role of scarcity in shaping value exchange behaviors [51]. Additionally, scarce items inherently attract increased attention, enhancing their cognitive significance [52]. This heightened perception of rarity not only reinforces the uniqueness of these items, but also inspires aspirations tied to identity and social status [31]. As a result, scarcity influences decision making and amplifies the prominence of rare items in social interactions.

Scarcity plays a pivotal role in consumer decision making, shaping how individuals process information and evaluate products [53]. Marketers frequently leverage scarcity to drive promotion strategies (see Appendix A), particularly through two common methods: "quantity--based scarcity" and "time-based scarcity." Emphasizing scarcity allows marketers to boost perceived value and drive purchase intentions [52, 54–56]. However, the effectiveness of scarcity strategies depends on various contextual factors, such as cultural and situational influences; for example, cultural norms and consumption contexts significantly shape consumer responses to scarcity [55,57]. Additionally, the inherent attributes of products—whether they are visually bold or more understated—can affect how scarcity is perceived and acted upon [50]. On social media, scarcity cues are prevalent, with limited-time trending topics or exclusive events often stimulating increased user engagement and content sharing [58]. Psychologists suggest that these behaviors reflect an intrinsic desire to share unique or limited information with others. Research further indicates that users are more likely to engage with scarce items that have attributes conducive to sharing, thereby amplifying the role of scarcity in social interactions [50]. These findings provide a robust theoretical basis for understanding how scarcity influences user behavior.

Although scarcity traditionally has been studied in conventional markets, its relevance to emerging markets, such as the NFT market, has become increasingly apparent. In the NFT context, scarcity is predominantly "attribute-based," where creators engineer scarcity by assigning probabilities to specific features [59]. Rarity scores, the primary metric for measuring NFT scarcity, effectively quantify this concept, making "rarity" and "scarcity" interchangeable in this domain [22]. Digital scarcity plays a crucial role in shaping NFT market dynamics. It affects such key aspects as pricing [29], market participation [21], and innovation [20]. Additionally, rare NFTs have significant social and symbolic value, particularly in signaling identity and status. This has driven the growing trend of giving NFTs as gifts [24,25,46].

However, the majority of existing research has predominantly concentrated on traditional scarcity strategies [36,60]. Conversely, attribute-based scarcity, as exemplified by NFTs, remains underexplored [61,62]. Within the digital ecosystem, scarcity not only shapes users' perceived value of NFTs, but also exerts a significant influence on their trading behaviors and decision making in social contexts. Digital scarcity is reshaping concepts of value and behavior in the NFT marketplace. Gaining deeper insights into its impact on users' perceptions and actions has become an essential area of inquiry.

3. Theoretical foundation and hypotheses

3.1. The relationship between digital scarcity and online behavior

In exploring the dynamics of the NFT market, digital scarcity has a pivotal role in shaping online trading behavior [3]. According to the scarcity principle, items perceived as rare are typically assigned higher value [4,63]. NFTs, as a unique class of digital assets, are deeply influenced by this principle. Their inherent scarcity often motivates users to adopt a long-term investment perspective, opting to retain these assets in anticipation of future value appreciation rather than engaging in frequent, short-term trading. This mirrors the behavior observed with traditional financial assets, where perceived rarity enhances an item's potential as an investment vehicle or collectible [41,64].

Highly scarce NFTs, valued for their uniqueness and potential for long-term appreciation, are commonly viewed as long-term holdings. Consequently, users are more inclined to hold these assets until favorable market conditions emerge, resulting in fewer transactions involving highly scarce NFTs [65]. This reluctance to trade reflects a strategic approach to maximizing returns, reinforcing the idea that higher levels of digital scarcity correlate with a reduction in trading frequency [4]. Thus, we propose the following hypothesis:

H1a. Digital scarcity of NFTs has a negative effect on users' online trading behaviors.

The concept of uniqueness is deeply embedded in human behavior, as individuals often strive to express their identity [66]. According to the scarcity principle, rare items not only hold greater value, but also enhance individuality and distinction [50,66]. This perception is particularly salient in the NFT market [54,67,68], where the psychological allure of scarcity drives users to participate in collecting and sharing activities to display their unique preferences and social standing [29].

In this context, NFTs carry significant exchange and symbolic value, offering both social status and a sense of belonging [44]. The rarity of NFTs heightens their appeal as gifts, providing emotional rewards by affirming the giver's social identity and enhancing the recipient's status within the community [22,69]. This dynamic has made NFTs particularly popular as gifts among younger generations [19,70]. Compared to traditional gifts, NFTs are viewed as more appealing because of their uniqueness, which stimulates gifting behaviors that enhance online community interactions. These exchanges amplify the psychological desire to share unique items or information, thereby strengthening social bonds and fostering community cohesion [60]. Moreover, these interactions contribute to the development of social capital and provide emotional satisfaction to users [59]. Accordingly, we propose the following hypothesis:

H1b. Digital scarcity of NFTs has a positive effect on users' online social behaviors.

3.2. The moderating effects of consistency of rarity evaluation

Uncertainty is a fundamental challenge in markets involving financial transactions, particularly in the volatile NFT landscape. As digital assets, NFTs inherently carry unpredictability that complicates their evaluation and trading [71]. To contextualize this challenge, traditional product sales rely on evaluations and certifications from trusted institutions to reduce uncertainty by signaling quality [72]. Similarly, in the NFT ecosystem, highly consistent rarity assessments across multiple platforms can diminish information asymmetry by providing unified quality signals. This alignment not only fosters confidence in digital scarcity [60,73], but also alleviates user uncertainty. Crucially, such consistency enhances the perceived credibility of evaluating platforms, thereby reinforcing market trust through reduced informational friction.

Reduced uncertainty enhances users' perception of NFT value. This perception fosters a broad market consensus, which elevates NFT valuations, amplifies their appreciation potential, and shapes investor decision making [70]. Highly consistent rarity evaluations reinforce users' acceptance of NFT valuations, solidifying expectations of future value appreciation [13]. Consequently, holders of high-rarity NFTs raise their minimum acceptable sale prices due to anticipated future returns [74]. Such expectations diminish the likelihood of transactions occurring at prevailing market prices, leading to a decline in the trading frequency of these NFTs. Thus, we formally hypothesize the following:

H2a. The consistency of rarity evaluation moderates the negative relationship between digital scarcity and online trading behavior. This negative relationship is significantly stronger when the consistency of rarity evaluation is high rather than low.

Sociological research highlights that individuals often showcase rare items to express their unique tastes and social status [74]. In the context of NFTs, when rarity evaluations are more consistent across platforms, users are more confident in the uniqueness and value of their NFTs. This consistent validation sends a positive signal to market participants, reinforcing the perception of NFTs as not just tradeable assets but also symbols of identity, taste, and social standing [75]. Consequently, the act of gifting or transferring high-rarity NFTs becomes more than a transactional exchange—it serves as an expression of social trust and reputation [76].

When validated through consistent rarity evaluations from independent platforms, high-rarity NFTs acquire demonstrable uniqueness, enhancing their social value. This motivates users to leverage such NFTs in social interactions to strengthen personal connections and demonstrate status. Moreover, highly consistent rarity evaluations reduce ambiguity about an NFT's rarity, clarifying its value in social settings [77]. This clarity fosters mutual understanding and facilitates the use of NFTs as tools for building and solidifying social networks. By mitigating uncertainty, consistent rarity evaluations heighten NFTs' desirability and social significance, enabling effective function as instruments of social exchange. Collectively, we hypothesize the following:

H2b. The consistency of rarity evaluation moderates the positive relationship between digital scarcity and online social behavior. This positive relationship is significantly stronger when the consistency of rarity evaluation is high rather than low.

3.3. The moderating effects of the impact of influencers

Social comparison literature underscores the influence of observing others on individual decision making and behavior [78]. This process enables individuals to reduce uncertainty by using external references to guide their choices [79]. In digital platforms, influencers—viewed as credible, knowledgeable, or trustworthy figures—frequently serve as key reference points for consumers [80]. Across such fields as marketing and investment, influencers play a significant role in shaping attitudes and behaviors [81]. In financial markets, individual investors often seek guidance from the decisions of more influential investors, particularly in the face of uncertainty. This reliance is even more pronounced in the digital economy, where influencers provide valuable insights for navigating the complexities of emerging financial instruments.

In the NFT ecosystem, digital scarcity is a key determinant of perceived value. As scarcity levels intensify, so too does the asset's appeal to users [64]. Nevertheless, heightened scarcity concurrently amplifies asset-related uncertainty, prompting more cautious decision-making among market participants [82]. In this study, influencers refer to individuals or institutions with substantial capital and market influence—commonly referred to as "whales" in the NFT ecosystem [83]. These whales are perceived as possessing expert knowledge of NFT investments [84]. When a collection attracts numerous whales, it signals potential market value, encouraging other investors to view it favorably. The perceived value of high-rarity NFTs leads investors to esteem them highly, anticipating future appreciation

driven by ongoing interest from whales. This expectation deters trading even amid short-term price changes, thereby aggravating the inverse relationship between rarity scores and transaction frequency.

Furthermore, the presence of numerous influencers bolsters market confidence in the stability of high-rarity NFTs. Users perceive these assets as less vulnerable to price fluctuations caused by smaller investors, fostering a belief in their resilience to risk [85]. This perception encourages long-term holding over frequent trading, as investors expect higher returns in a stable market environment. The presence of whales thus further aggravates the negative relationship between digital scarcity and trading frequency. Therefore, we propose the following hypothesis:

H3a. The impact of influencer moderates the negative relationship between digital scarcity and online trading behavior. This negative relationship is significantly more pronounced when the impact of influencer is strong rather than weak.

In collections of high-rarity NFTs, the involvement of whales plays a crucial role in enhancing the appeal of these assets. When a significant number of whales actively participate in projects focused on high-rarity NFTs, their actions substantially increase the visibility and social recognition of these items [83]. This heightened visibility in turn links high-rarity NFTs to symbolic qualities such as status, identity, and exclusivity [86]. As a result, users are more likely to perceive high-rarity NFTs not just as investment assets, but also as meaningful and prestigious social symbols [50].

Moreover, whale participation amplifies both the visibility and symbolic value of high-rarity NFTs, further strengthening the relationship between their rarity and their desirability as social gifts [87]. This dynamic encourages users to present high-rarity NFTs as special social tokens, thereby increasing the frequency of their social transfer and reinforcing the positive correlation between rarity and social interactions.

Additionally, whales often serve as focal points of discussion within the NFT market. In collections with strong whale involvement, whales' activities—such as public endorsements or discussions—create heightened visibility and recognition, making these NFTs more prominent and socially significant. This elevated relevance motivates users to transfer high-rarity NFTs more often in social contexts, as such exchanges are seen as a way to enhance social interactions, build prestige, and foster deeper engagement [88]. Thus, we propose the following hypothesis:

H3b. The impact of influencer moderates the positive relationship between digital scarcity and online social behavior. This positive relationship is significantly more pronounced when the impact of influencer is strong rather than weak.

The research model in Fig. 1 depicts the hypothesized relationships.

4. Methodology

This study focuses on NFTGo, a prominent platform in the NFT ecosystem, as the primary data source. First, NFTGo offers extensive and diverse data on NFTs, encompassing major transactions across multiple blockchains and various asset types. Its broad coverage and advanced search capabilities make it a rich repository of high-quality market data, providing a solid foundation for our analysis. Second, the platform's data is representative of global NFT market dynamics, aggregating information from multiple marketplaces and creators. This diversity aligns with our objective of examining trends and patterns across the broader NFT landscape, thereby increasing the relevance and applicability of our findings. NFTGo also adheres to rigorous data collection and processing standards, ensuring accuracy, completeness, and reliability. Its robust visualization tools and market metrics support in-depth analysis, strengthening the validity of our conclusions.

4.1. Data

The dataset used in this study comprises detailed information on a total of 866,340 NFTs from the top 100 collections listed on the homepage of the NFTGo platform, collected between November 24, 2023, and November 30, 2023. This selection was guided by several key considerations. First, these top-ranked collections exert substantial influence within the market. Their trading data and user behavior patterns not only serve as a reliable representation of the broader NFT market ecosystem, but also more accurately capture overarching market trends. As such, they provide a robust foundation for validating our theoretical framework. Second, these highly visible collections typically have more robust and comprehensive data records, ensuring a rich foundation for analysis. Finally, by focusing on collections of similar market rankings, this approach helps control extraneous variables that otherwise might influence the results. This enables a more precise and focused exploration of the causal relationship between digital scarcity and user behavior.

To support our analysis, the dataset includes a range of metrics for each NFT, including rarity scores, sales counts, transfer counts, whale numbers, and comparative rarity evaluations from other platforms. These metrics form the cornerstone of our study. After the initial data collection, a thorough data cleansing process was undertaken. Skewed distributions were adjusted using logarithmic transformations, and data

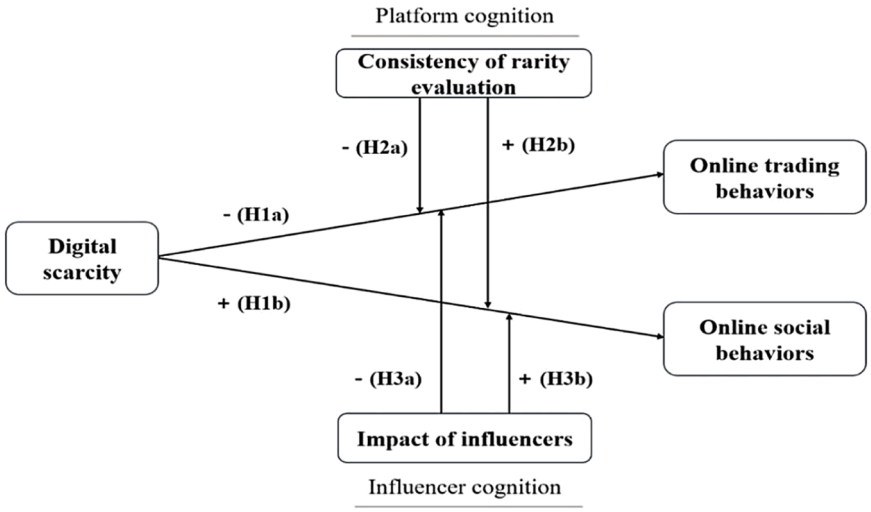


Fig. 1. Research model.

formats unsuitable for analysis were standardized. Additionally, records with missing values in critical fields were excluded, resulting in the removal of 17,211 observations. Consequently, the final dataset for analysis includes 849,129 NFTs.

Within the NFT ecosystem, wash trading demands attention due to its potential to distort value perception. This practice artificially inflates NFT values through repetitive sales/transfers between linked wallets [27]. It is important to note that wash trading encompasses not only sales, but also transfers. However, this study focuses on the influence of rarity on user behaviors rather than on transactional patterns. Moreover, the NFTGo platform uses advanced detection mechanisms to flag suspicious activity, maintaining data integrity and mitigating fraud risks. Given these safeguards and our behavioral focus, wash trading does not compromise this study's validity.

4.2. Variables

Dependent variables. There are two dependent variables in our model: users' *online trading behaviors* (*logTrading*) and *online social behaviors* (*logSocial*). The NFTGo platform offers comprehensive activity records for each NFT, detailing the number of trades and transfers. Currently, many NFT platforms provide a "transfer" feature, enabling users to send desired NFTs to the recipient's wallet address with minimal effort. In this context, "transfer" refers to nonmonetary transactions, often conducted for social purposes. In our study, we represent online trading behaviors by the total number of an NFT's sales up to the time of data collection, which means that when a user sells the NFT to another user, this is recorded as a trade [13]. Meanwhile, the number of NFT transfers represents users' online social behaviors. Typically, individuals often transfer NFTs as gifts to others for the purpose of social recognition or interpersonal interaction [45]. To reduce skewness, we take the logarithm of the dependent variables.

Independent variables. *Digital scarcity*, as noted previously, is typically characterized by rarity scores based on certain features. Hence, we use the rarity scores of NFTs provided by the platform to concretely measure digital scarcity. The platform calculates a rarity score for each NFT using the Jaccard distance algorithm. The Jaccard distance measures the dissimilarity between sample sets on a scale from 0 (complete similarity) to 1 (complete dissimilarity). Specifically, for two NFT sets A and B, the Jaccard distance is calculated using the formula $J(A, B) = 1 - \frac{|A \cap B|}{|A \cup B|}$, where $|A \cap B|$ is the number of elements common to both sets, and $|A \cup B|$ is the total number of elements in either set.

To calculate the rarity score of an NFT, first determine the Jaccard distance between the target NFT and all other NFTs in the collection. The average of these distances is then computed to obtain initial data for the rarity score. This data is processed further using normalization, specifically the Z-score normalization, calculated as $Z_i = \frac{x_i - \min(x)}{\max(x) - \min(x)}$. The resulting Z-score is multiplied by 100 to yield a final rarity score, which falls within a range of 0 to 100; the higher the rarity score, the more unique the NFT. Compared with other methods of calculating rarity score, this offers more reasonable results for calculating rarity score. This method emphasizes the data aspects of an NFT and ensures that the process is objective and transparent. For analysis, we normalize this score by dividing it by 100, which serves as our independent variable [89].

Moderating variables. The *consistency of rarity evaluation* is measured by comparing the NFT rarity rankings from three platforms. On the NFTGo platform, in addition to its own rankings, it also presents rankings from two other notable trading platforms (Rarible.com and Opensea.com). The consistency of these rankings across different platforms is crucial for participants in assessing the value of an NFT. Following Héberger's sum of ranking differences method [90], we calculate the pairwise differences in rankings across the three platforms: P1, P2, and P3. These differences are summed and divided by 3 to derive a continuous variable, the mean rank difference. The formula is as follows: $(|P1\text{-}P2| + |P1\text{-}P3| + |P2\text{-}P3|)/3$. A smaller difference indicates greater consistency, while a larger difference indicates less consistency. To enhance interpretability, we invert this variable so that higher values indicate greater consistency and lower values indicate less consistency. To reduce skewness, we take the logarithm of this variable.

The *impact of influencers* is measured by the number of influencers involved in the project. Here an influencer refers to an individual or institution with substantial assets, often called "whales" in the NFT space. These influencers possess significant wealth and visibility, enabling them to better understand market trends and influence other users' decisions. Therefore, for the purposes of this study, we use this measure to assess the impact of influencers [91]. We take the logarithm of this variable to reduce skewness. The description and descriptive statistics of the variables are shown in Table 2.

4.3. Empirical models

In line with our proposed hypotheses, we posit that digital scarcity (*Scarcity*) negatively affects users' online trading behaviors (*logTrading*) while positively affecting their online social behaviors (*logSocial*). Additionally, it is crucial to consider the potential moderating effects of the consistency of rarity evaluation (*Evaluation*) and the impact of influencers (*logInfluencer*) as significant factors that may influence user behaviors within the NFT market. Therefore, we have incorporated these factors into our research model. The impacts of digital scarcity on online trading and social behaviors, along with the corresponding moderating effects, are presented in Eqs. 1 and 2, respectively.

$$\begin{aligned} \textit{Online trading behavior} = {} & \alpha_0 + \alpha_1 \textit{digital scarcity} \\ & + \alpha_2 \textit{consistency of rarity evaluation} \\ & + \alpha_3 \textit{peer influence} + \alpha_4 \textit{digital scarcity} \\ & * \textit{consistency of rarity evaluation} \\ & + \alpha_5 \textit{digital scarcity} * \textit{impact of influencers} \\ & + \alpha_{controls} \textit{controls} + \varepsilon \end{aligned} \tag{1}$$

$$\begin{aligned} \textit{Online social behavior} = {} & \beta_0 + \beta_1 \textit{digital scarcity} \\ & + \beta_2 \textit{consistency of rarity evaluation} \\ & + \beta_3 \textit{peer influence} + \beta_4 \textit{digital scarcity} \\ & * \textit{consistency of rarity evaluation} \\ & + \beta_5 \textit{digital scarcity} * \textit{impact of influencers} \\ & + \beta_{controls} \textit{controls} + \varepsilon \end{aligned} \tag{2}$$

Here *controls* represent the control variables that account for additional factors potentially influencing user behavior, ensuring that the results are robust and not confounded by these extraneous variables. ε denotes the error term, capturing unobserved factors and the inherent variability in behavior that the model does not explicitly account for.

We incorporate control variables to account for potential confounding factors that may influence users' online behaviors in the NFT market. First, we include the lifespan of the NFT (*logDuration*), defined as the number of days from its minting to the point of data collection. This variable is included to account for the impact of varying lifespans on the number of trades and transfers undergone by an NFT. Next, recognizing the potential influence of collection ranking on user behaviors within the NFT market, we included the ranking of NFT collections (*logRanking*) as a significant control variable in our model. Finally, we include the number of inherent traits (*logTraits*) and the average sale price (*logPrice*) of the NFT to improve the robustness of our analysis. We then applied a natural logarithmic transformation to these variables before estimation to reduce skewness.

Before conducting the analysis, we examine relations and collinearity among the variables. Pearson correlation outcomes (Table 3) revealed significant correlations between core variables. The maximum

Table 2
Definition and descriptive statistics of the main variables (N = 849,129).

	Variable	Description	Mean	S.D.	Min	Max
Independent variable	Digital scarcity	The rarity score of each NFT.	38.041	24.478	0.000	100
Dependent variables	Online trading behaviors	The total number of times the NFT has been traded.	1.194	4.159	0.000	1492
	Online social behaviors	The total number of transfers without money exchanged for the NFT.	1.199	5.804	0.000	1207
Moderating variables	Consistency of rarity evaluation	The consistency of the rarity ranking for the same NFT across three platforms.	1078.443	1641.964	0.000	10890
	Impact of influencers	The number of people who holds substantial amounts of NFTs.	84.550	122.187	0.000	702
Control variables	Duration	The time elapsed since the creation of the NFT.	725.858	223.125	182.000	2311
	Sale price	The average sale price of the NFT.	18337.240	2074481.000	0.000	6.00e+08
	Characteristics	The number of traits of the NFT.	7.614	4.671	0.000	43
	Quality	The ranking of the collections to which the NFT belongs.	565.603	649.441	1.000	2381

Table 3
Pearson correlation matrix.

		1	2	3	4	5	6	7	8
1	*Scarcity*	1							
2	*logTrading*	-0.020***	1						
3	*logSocial*	0.023***	0.583***	1					
4	*logEvaluation*	-0.024***	0.002*	-0.014***	1				
5	*logInfluencer*	-0.035***	0.105***	-0.014***	0.123***	1			
6	*logDuration*	0.049***	0.378***	0.376***	-0.041***	0.077***	1		
7	*logPrice*	0.108***	0.113***	0.043***	0.062***	0.103***	0.020***	1	
8	*logTraits*	0.199***	0.000	-0.036***	0.024***	0.114***	0.003***	0.002	1
9	*logRanking*	0.003**	-0.058***	0.046***	-0.066***	-0.753***	0.058***	-0.096***	-0.059***

Note: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

variance inflation factor (VIF) recorded for the variables was 1.10, with an average of 1.05, both substantially below the accepted collinearity threshold of 10, indicating negligible collinearity within the dataset.

5. Results

5.1. Main hypothesis tests

Table 4 presents the results of the empirical analysis. We first estimated Models 1 and 4 with only the control variables and the two moderating variables, after which we added the main explanatory variables. Regarding the main effect, Model 2 indicates that digital scarcity negatively impacts online trading behaviors (α_1= -0.132, $p <$ 0.001), thus providing support for H1a. In addition, Model 5 indicates that digital scarcity positively impacts online social behaviors (β_1 = 0.007, $p < 0.01$), thus supporting H1b.

The moderating effects of the consistency of rarity evaluation and impact of influencers are shown in Models 3 and 6. The findings suggest a negative impact of the consistency of rarity evaluation on the influence of digital scarcity on online trading behaviors (α_4 = -0.033, $p < 0.001$), thus supporting H2a. At the same time, the results reveal a positive impact of the influence of digital scarcity on online social behaviors

Table 4
Regression results of the main analysis.

Variables	*logTrading*			*logSocial*		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Scarcity logInfluencer*			0.107***			0.160***
			(0.002)			(0.002)
Scarcity logEvaluation*			-0.033***			0.016***
			(0.001)			(0.001)
Scarcity		-0.132***	-0.127***		0.007**	0.022***
		(0.003)	(0.003)		(0.003)	(0.003)
logInfluencer	0.016***	0.013***	0.010***	-0.031***	-0.028***	-0.025***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
logEvaluation	0.001***	0.001***	0.014***	0.001***	0.002***	-0.004***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
logDuration	1.055***	1.063***	1.066***	1.019***	1.012***	1.003***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
logPrice	0.103***	0.107***	0.104***	0.040***	0.040***	0.041***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
logTraits	-0.009***	0.005***	-0.003**	-0.038***	-0.025***	-0.030***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
logRanking	-0.019***	-0.021***	-0.021***	-0.006***	-0.006***	-0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Constant	-6.492***	-6.500***	-6.451***	-6.103***	-6.099***	-6.091***
	(0.019)	(0.019)	(0.018)	(0.018)	(0.018)	(0.018)
N	849,129	849,129	849,129	849,129	849,129	849,129
R^2	0.159	0.162	0.166	0.146	0.145	0.150

Notes: Standard errors in parentheses, *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

($\beta_4 = 0.016$, $p < 0.001$), thus lending support to H2b.

In terms of the impact of influencers, we find that it positively influences the effect of digital scarcity on both online trading behaviors ($\alpha_5 = 0.107$, $p < 0.001$) and online social behaviors ($\beta_5 = 0.160$, $p < 0.001$), thus supporting H3b. However, the results do not support H3a, which proposed a negative moderating effect of influencers on the association between digital scarcity and online trading behaviors. This somewhat unexpected result may be attributed to the presence of numerous influencers, which enhances the exposure and popularity of NFTs. This in turn attracts more potential buyers to the market and boosts confidence in the trading sector. Moreover, influencers act as significant market signals, greatly impacting other users' perceptions of scarcity. This influence encourages a wider user base to participate more actively in NFT trading, with individuals more likely to trade frequently rather than adopting a passive long-term holding strategy in anticipation of future appreciation. We summarize these findings in Table 5.

To enhance the intuitive understanding of the moderating roles of the platform and influencers—two key subjects—we construct diagrams to illustrate the effects of these moderating variables at various levels. Specifically, Figs. 2 and 3 elucidate the moderating role of the consistency of rarity evaluation, while Figs. 4 and 5 depict the moderating role of the impact of influencers.

The negative impact of digital scarcity on users' online trading behaviors is more pronounced when the consistency of rarity evaluation is high compared to scenarios with low consistency, as shown in Fig. 2. Conversely, Fig. 3 illustrates that with a high consistency of rarity evaluation, digital scarcity has a stronger positive influence on users' online social behaviors compared to scenarios of low consistency. These findings indicate that the consistency in platforms' evaluations of rarity serves as a crucial boundary condition. When rarity evaluations demonstrate high consistency, they generate standardized valuation signals that propagate through user networks. This signal diffusion not only elevates perceived scarcity premiums but also cognitively anchors uniqueness attributions among collectors. Consequently, this will simultaneously strengthen digital scarcity's inhibiting effect on trading behaviors and amplify its promoting effect on social behaviors.

As shown in Fig. 4, when the impact of influencers is strong, the negative effect of digital scarcity on online trading behaviors is positively moderated and more pronounced than in situations with lower levels of impact of influencers. This suggests the robust role of influencers, which can even counteract the adverse effects that digital scarcity may have on trading activities. This observation differs from H3a,

Table 5
Summary of findings.

Hypotheses		Supported
H1a	Digital scarcity of NFTs has a negative effect on users' online trading behaviors.	Yes
H1b	Digital scarcity of NFTs has a positive effect on users' online social behaviors.	Yes
H2a	The consistency of rarity evaluation moderates the negative relationship between digital scarcity and online trading behavior. This negative relationship is significantly stronger when the consistency of rarity evaluation is high rather than low.	Yes
H2b	The consistency of rarity evaluation moderates the positive relationship between digital scarcity and online social behavior. This positive relationship is significantly stronger when the consistency of rarity evaluation is high rather than low.	Yes
H3a	The impact of influencer moderates the negative relationship between digital scarcity and online trading behavior. This negative relationship is significantly more pronounced when the impact of influencer is strong rather than weak.	No
H3b	The impact of influencer moderates the positive relationship between digital scarcity and online social behavior. This positive relationship is significantly more pronounced when the impact of influencer is strong rather than weak.	Yes

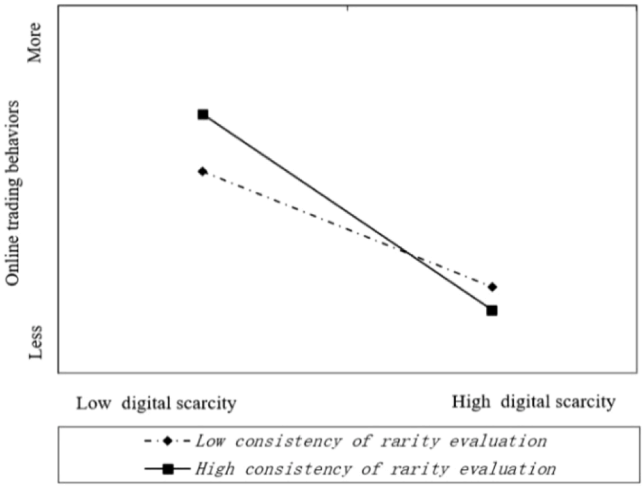


Fig. 2. The moderating effect of the consistency of rarity evaluation on trading behaviors.

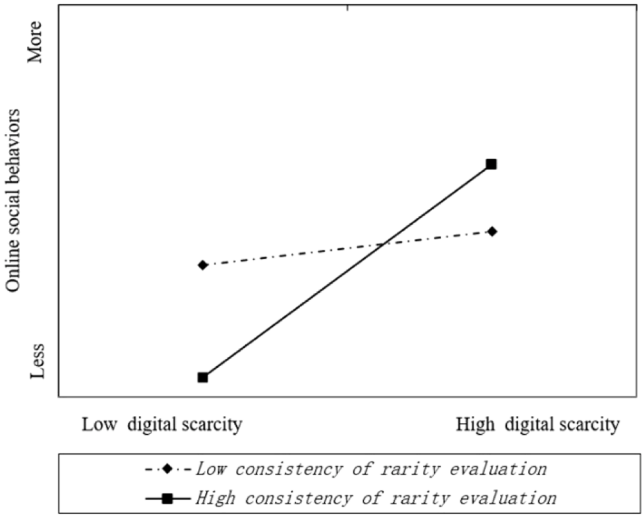


Fig. 3. The moderating effect of the consistency of rarity evaluation on social behaviors.

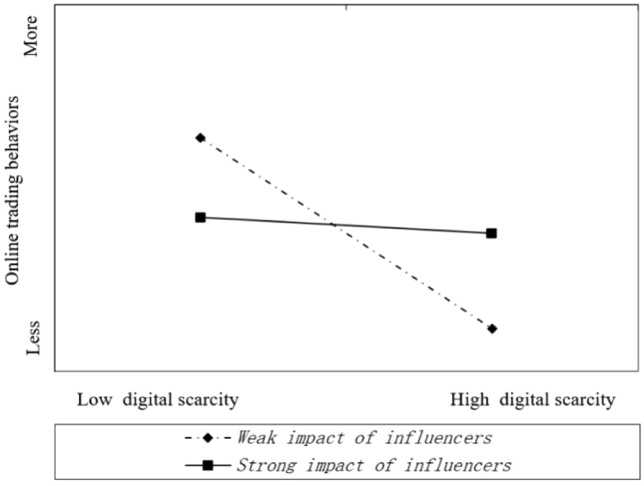


Fig. 4. The moderating effect of the impact of influencers on trading behaviors.

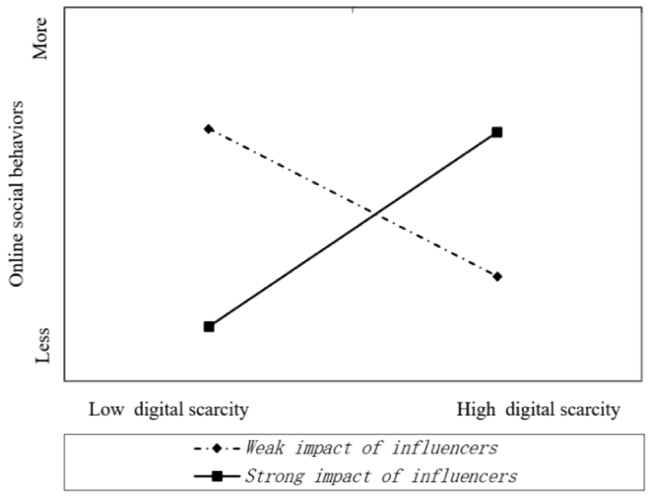


Fig. 5. The moderating effect of the impact of influencers on social behaviors.

and as mentioned previously, we speculate that the discrepancy between our hypothesis and empirical results may stem from the significant influence exerted by influencers within this market. Their presence can sway users' attention and confidence, thereby enhancing market activity. In this context, a stronger influencer impact signifies greater recognition for the NFT, which might heighten interest in the NFT and increase the likelihood of transactions, consequently boosting trading volume.

The positive impact of digital scarcity on online social behaviors is markedly more pronounced in the context of strong influencer impact compared to weak influencer impact, as illustrated in Fig. 5. This finding indicates that the impact of influencers functions as a pivotal indicator, emitting positive signals that not only counteract the adverse effects of digital scarcity on trading behaviors, but also intensify the positive repercussions on social interactions. Even within the digital sphere, where digital goods are traded, the actions of influencers provide individuals with valuable benchmarks. Given the inherent virtuality and elevated risk associated with the digital domain, the role of influencers may indeed be more significant here than in conventional offline contexts.

5.2. Endogeneity and identification

In this study, the issue of reverse causality does not arise because the circulation details of each NFT are publicly available prior to their release, and their rarity remains constant over time [3]. Also, measurement error is not a concern, as the primary variables are derived from existing platform data. However, the presence of certain unobservable factors, which cannot be included as control variables in the model, may lead to omitted variable bias and result in endogeneity issues [92]. Additionally, self-selection bias may introduce further bias into our estimates [93]. Therefore, we explore solutions to address potential endogeneity concerns below.

We use the control function approach [94,95], which incorporates additional variables (known as "instruments") into the empirical model to account for the influence of unobserved factors related to the endogenous variable. If effectively implemented, the remaining variation in the endogenous variable will be independent of the error term. By considering the impact of these additional variables on the rarity score as a dependent variable in the selection regression equation, the rarity score as an independent variable is no longer related to the error term in the regression equations (Eqs. 1 and 2). This adjustment allows us to establish the assumption of independence between the rarity score and the error term.

Following the method outlined by Petrin and Train [94], we performed an estimation using an exogenous (excluded) variable as a covariate. This variable affects the endogenous variable (digital scarcity) but is not related to either trading behaviors or social behaviors. The predicted residuals from this estimation provide effective control variables to address endogeneity concerns. We used "the number of NFTs in a collection" because this variable is highly related to the rarity score, thereby meeting the relevance condition. Importantly, there is no evidence that it is directly associated with users' online behaviors, fulfilling the exogeneity condition. After performing the estimation (see Appendix B), we incorporated the predicted residuals into the performance equations (Eqs. 1 and 2) to control for self-selection concerns.

5.3. Robustness check

We used a variety of alternative specifications to test the robustness of the aforementioned results, which have proven to be consistent. Specifically, we conducted three types of robustness checks to evaluate our findings. First, we used alternative models, including heteroscedasticity-robust regression and Poisson regression, to triangulate the results. Second, we substituted the independent variable with a measure of rarity ranking to assess the outcomes. Third, we incorporated additional control variables into the regression model.

Columns (1) and (3) in Table 6 report the results of the heteroscedasticity-robust regression, indicating that the obtained results are still robust even after considering the heteroscedasticity problem. Moreover, given that the dependent variable data used in this paper are nonnegative integers representing counts, the variance is equal to the mean, which aligns with the data prerequisites for Poisson regression [96]. Therefore, we subjected our research findings to further verification through Poisson regression, with results presented in columns (2) and (4). The data clearly substantiate the negative impact of digital scarcity on online trading behaviors and its positive influence on online social interactions. The moderating effects are consistent with our previous analysis.

To further test the robustness of our findings, we replaced rarity scores with rarity rankings to quantify the rarity of NFTs [13]. The results remained consistent (see Appendix C). Because a smaller ranking difference indicates higher rarity, which is completely opposite to the meaning of rarity represented by the independent variable rarity score that we used earlier, we negated the ranking prior to analysis to better align its inherent meaning of rarity, thereby facilitating a more accurate understanding of the results. Additionally, we included as many control variables as possible that potentially could affect user behaviors [97], including NFT type (*Type*), number of collection traits (*logPtraits*), number of collectors (*logSave*), and number of powerful holders (*logQualityholders*). The results remained robust (see Appendix D).

In conclusion, the findings of this study demonstrated robustness across various measurement models irrespective of whether we used rarity scores, rarity rankings, or alternative evaluative frameworks. Specifically, these tests substantiated our hypothesis that digital scarcity significantly influences users' online behaviors within the NFT digital asset marketplace. Our research offers valuable insights into how users assess the value of NFTs and provides useful information for market participants in formulating their strategies.

6. Discussion

6.1. Key findings

Our econometric analyses explore the effects of digital scarcity on user behavior in the NFT market, with a particular focus on the moderating roles of platforms and influencers. First, our findings reveal digital scarcity's dual role in NFT markets, contextualized within evolving market dynamics. The paradoxical relationship, where scarce NFTs exhibit reduced trading frequency but increased transfer activity, is a notable aspect of the market. This duality highlights the multifaceted nature of NFTs as tools for both wealth accumulation and community

Table 6
Heteroscedasticity-robust regression and Poisson regression.

Variables	*logTrading*		*logSocial*	
	(1) Heteroscedasticity-Robust	(2) Poisson Regression	(3) Heteroscedasticity-Robust	(4) Poisson Regression
Scarcity logInfluencer*	0.107***	0.257***	0.160***	0.409***
	(0.002)	(0.006)	(0.002)	(0.007)
Scarcity logEvaluation*	-0.033***	-0.132***	0.016***	0.008***
	(0.001)	(0.002)	(0.001)	(0.002)
Scarcity	-0.127***	-0.401***	0.022***	0.124***
	(0.003)	(0.008)	(0.003)	(0.008)
logInfluencer	0.010***	-0.012***	-0.025***	-0.103***
	(0.001)	(0.002)	(0.001)	(0.003)
logEvaluation	0.014***	0.042***	-0.004***	-0.005***
	(0.000)	(0.001)	(0.000)	(0.001)
logDuration	1.066***	1.727***	1.003***	1.794***
	(0.005)	(0.006)	(0.004)	(0.005)
logPrice	0.104***	0.188***	0.041***	0.071***
	(0.001)	(0.002)	(0.001)	(0.003)
logTraits	-0.003***	0.105***	-0.030***	-0.012***
	(0.001)	(0.003)	(0.001)	(0.003)
logRanking	-0.021***	-0.067***	-0.001	-0.008***
	(0.001)	(0.002)	(0.001)	(0.002)
Constant	-6.451***	-11.934***	-6.091***	-12.531***
	(0.029)	(0.037)	(0.025)	(0.035)
N	849,129	849,129	849,129	849,129
R^2	0.166		0.150	

Note: Robust standard errors in parentheses, *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

engagement.

Second, the consistency of rarity evaluations across platforms amplifies this duality. High consistency in rarity evaluations negatively moderates the relationship between digital scarcity and trading behaviors but positively moderates its connection with social behaviors. This suggests that when rarity evaluations are highly consistent across different platforms, users may develop stronger expectations of future value appreciation, prompting them to retain rare NFTs rather than sell them immediately, thereby reducing trading frequency. At the same time, the broadly recognized status of rarity may increase the desirability of these NFTs, enhancing their role in fostering connections and interactions within the digital asset community.

Finally, we emphasize the pivotal role of influencers in shaping NFT market behavior. Influencers positively impact the relationship between digital scarcity and both trading activity and social interactions. Consistent with prior research on influencer marketing, our results demonstrate that influencers enhance the visibility and desirability of scarce NFTs, driving both market participation and social engagement [98–100]. Notably, contrary to our initial hypothesis, influencers facilitate, rather than suppress, trading activity involving scarce NFTs, providing a potential solution to the issue of declining transactions in the NFT market. These findings extend the existing literature by integrating the concept of scarcity with the behavioral aspects of digital assets, offering a nuanced understanding of how market forces, social influence, and digital scarcity interact.

6.2. Theoretical contributions

Our study makes several contributions to the academic literature. First, it advances existing NFT research by addressing the paradoxes of digital scarcity in relation to user behaviors. Unlike prior studies, which focused primarily on the economic outcomes of digital scarcity [29,45,48], our study adopts a behavior-oriented process perspective, examining how scarcity influences both trading and social interactions on NFT platforms. To the best of our knowledge, this is the first study to explore the role of digital scarcity in shaping social behaviors within the NFT ecosystem. By highlighting the differential effects of digital scarcity on user trading and social activities, our findings provide valuable insights for the design and strategic development of NFTs.

Second, our study deepens the theoretical understanding of the scarcity principle as it applies to digital ecosystems by focusing on digital goods like NFTs. Traditional markets often associate greater scarcity with increased trading frequency [31,50]; however, our findings reveal a counterintuitive dynamic in the NFT market: increased scarcity reduces trading activity. This divergence can be attributed to the investment-oriented nature of NFTs, where scarcity increases perceived value, motivating users to hold onto these assets rather than trade them. This insight challenges conventional assumptions about scarcity and trading behavior, broadening the applicability of the scarcity principle to digital environments. Additionally, our work contributes to the broader discourse in information systems research by offering a nuanced understanding of scarcity in the context of digital goods.

Third, our research expands theoretical frameworks related to NFTs by introducing two critical boundary conditions that influence user behavior under conditions of uncertainty: the consistency of rarity evaluations across multiple platforms and the impact of influencers. Consistency in rarity assessments serves as a critical moderator that accentuates digital scarcity's dual behavioral outcomes. High consensus reinforces suppressed trading activity by validating perceived value, while simultaneously enhancing social engagement via amplified status-signaling utility. This moderating function transcends informational value, actively potentiating behavioral responses to digital scarcity.

Moreover, our study highlights the pivotal role of influencers in shaping user behavior in the NFT market—an aspect largely overlooked in prior research. Specifically, our findings demonstrate that influencers significantly enhance the visibility and perceived desirability of scarce NFTs, thereby reshaping trading dynamics. This offers a fresh perspective on the dynamics of information dissemination and decision making in socially interactive digital markets. By demonstrating the influence of social comparison psychology and the critical role of influencers as key nodes in virtual social networks, our research contributes to the expanding literature on influencer marketing and its intersection with user behavior in digital asset ecosystems.

6.3. Managerial implications

6.3.1. Practical implications for NFT ecosystems

For NFT creators (designers), our findings highlight the strategic

importance of digital scarcity. By emphasizing NFTs' uniqueness, creators can enhance their appeal, stimulate social interactions, and foster user engagement [23]. However, overemphasizing scarcity may restrict circulation and dampen trading activity. To balance uniqueness and liquidity, creators should judiciously apply scarcity mechanisms while leveraging influencers to amplify users' perception of scarcity. Our results show that influencer presence mitigates the negative relationship between digital scarcity and trading behavior, suggesting that collaborations with influencers can drive market participates' interest and transaction activity.

For platform managers, our results suggest that influencers can mitigate the adverse effects of digital scarcity on trading behavior. High-profile users, or whales, can bolster user confidence in rare NFTs and stimulate trading. Platforms should strategically leverage influencers by providing tools to visualize their activities (e.g., holding periods, trading histories, and transfer patterns), thereby enhancing overall market transparency. By providing users with more comprehensive and high-quality insights into influencer behavior, platforms can effectively reduce the uncertainty users face in decision making processes, thus building trust in the platform and transactions, while encouraging users to make more informed decisions. This not only aids users by better market participation, but also fosters healthy and orderly market development.

Our research further emphasizes the importance of consistent rarity evaluation. It can reduce information asymmetry, enhance users' perception of scarcity, and encourage the use of rare NFTs in social activities. Managers should consider enhancing the evaluation of NFT rarity in areas related to social interactions, providing valuable reference points for users. Additionally, by improving the reliability and transparency of these evaluations—such as by disclosing the rarity assessment process—they can build greater user trust, foster deeper social connections, and increase overall engagement. Ultimately, such consistency lays the groundwork for a more dynamic and sustainable NFT ecosystem.

The NFT market's dynamics provide transferable strategies for token ecosystems, including emerging assets like Meme coins and AI tokens. Creators and managers of these assets can focus on managing digital scarcity and leveraging influencers. Through scientific scarcity design and comprehensive insights into the behavior of influencers, they can further enhance the vitality and liquidity of the asset market.

6.3.2. *Ecosystem governance: mitigating risks in crypto markets*

This study also offers actionable governance strategies for addressing systemic risks (e.g., hypothetical pricing, collusion among buyers and sellers, artificially imposed scarcity) in cryptocurrency markets [101]. First, prior studies have demonstrated that NFT scarcity positively influences transaction prices [100]. This research further reveals that scarcity reduces trading frequency. Building on these findings, we propose integrating trading frequency and asset turnover rates into NFT valuation models to enhance pricing mechanisms. Specifically, this approach dynamically captures shifts in market liquidity (as opposed to relying solely on static scarcity metrics). For instance, high trading frequency coupled with low asset turnover may signal speculation-driven hypothetical pricing, whereas low trading frequency with high turnover could reflect value consensus. By anchoring valuations to real-time on-chain transaction behaviors, it can mitigate distortions caused by short-term speculation.

Second, the prices of NFTs can be easily inflated through artificial trading among beneficiary-controlled wallets [27]. Institutions or large investors with stronger financial capabilities are more capable of carrying out such operations. Our study reveals that influencers play a crucial role in shaping the relationship between digital scarcity and user behavior, requiring that policymakers and platforms strategically leverage this influence. Beyond detecting synchronized trading patterns (transaction homogeneity) via network analysis, it is advisable to consider establishing an on-chain reputation mechanism that links transaction transparency to reputation, thereby encouraging influencers to transition from arbitrage facilitators to responsible advocates.

Third, the moderating effect of cross-platform scarcity evaluation consistency demonstrates that multisource validation mechanisms can significantly enhance market trust in perceived digital scarcity. This finding necessitates establishing a more objective, multidimensional verification framework for crypto asset scarcity to mitigate excessive artificial scarcity. For example, integrating technical scarcity, community-driven scarcity, and algorithmic scarcity instead of relying on a single scarcity evaluation method or the metrics of a single platform can enhance the scientificity and rigor of artificial scarcity.

6.4. *Limitations and future research*

Although this study offers valuable insights, it has certain limitations that suggest avenues for future research. One primary limitation is its narrow focus on a specific subset of data, exclusively sourced from the NFTGo platform, which specializes in providing NFT market analytics. Despite yielding robust findings, our study is limited by the lack of a broader cross-platform analysis of NFT scarcity. Therefore, future research should aim to provide a more extensive and detailed investigation into the relationship between digital scarcity and user behaviors.

Another limitation of this study lies in the methodology used to quantify digital scarcity. We relied on a platform-generated rarity score that evaluates the uniqueness of NFT features using Jaccard distance. This method has its own limitations and potential biases. A notable limitation is the assumption that all features are of equal importance, whereas in the NFT market, certain features may have significantly more influence on determining rarity. Furthermore, this method does not account for the informativeness of rare features compared to common ones, nor does it consider the frequency of features within the collection. Additionally, perceptions of uniqueness and desirability in the NFT market are inherently subjective and might not always align with algorithmic assessments. However, advancements in computer vision and machine learning present transformative opportunities to address these gaps. Future studies could leverage these technologies to analyze the visual and design elements that users find most engaging. Such insights could not only refine the measurement of digital scarcity, but also guide the creation of NFTs that align more closely with user preferences.

CRediT authorship contribution statement

Hua Lu: Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Xin (Robert) Luo:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Data curation. **Shuang (Sara) Ma:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition. **Runyu Chen:** Writing – review & editing, Resources, Project administration, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendices

Appendix A. Literature on scarcity

Author	Context	Virtual context	Type of scarcity	Moderators	Dependent variable	Conclusion
Hughes, 2022	Collectible card game	×	Quantity-based scarcity	——	Price	The odds of obtaining manufacturer created rarity positively affect prices.
Li et al., 2021	Hospitality businesses	×	Quantity-based scarcity	——	Purchase Intention	In the context of the pandemic, consumers perceive scarce hotel services as having lower safety in terms of consumption. As a result, this reduces their willingness to purchase and preference.
Song et al.,2021	Hospitality businesses	×	Quantity-based scarcity and time-based scarcity	Social cue messages	Willingness to purchase	Presenting social cue messages enhances perceived consumer competition and consequently purchase intention when a limited quantity scarcity message is presented.
Wu et al., 2021	Milk tea	×	Quantity-based scarcity and time-based scarcity	Personal impulsiveness	Impulse purchase	Limited-quantity scarcity and limited-time scarcity increased a consumer's perceived arousal, which then led to impulse purchase.
Huang et al.,2020	Restaurant booking	×	Quantity-based scarcity	Consumer's sense of power	Purchase Intention	The demand-framed (vs. supply-framed) scarcity appeal leads to higher purchase intention among consumers with a high sense of power.
Noone and Lin,2020	Hospitality businesses	×	Quantity-based scarcity and time-based scarcity	Booking lead-time	Booking Intention	Limited-time scarcity and limited-quantity scarcity are effective in leading to a positive effect on booking intentions when booking lead time is long.
Cachon et al.,2018	Automobile sale	×	Quantity-based scarcity	——	Sales	If the vehicles added to the inventory expand the available sub-model groups, then sales will increase. However, if the added vehicles belong to the same sub-model as the existing ones, sales will actually decrease.
Kapferer and Valette-Florence,2018	Luxury sale	×	——	Cultural differences	Brand desirability	The overall effect of increased market penetration on luxury desirability is negative.
Lee et al.,2018	Sunglasses	×	Quantity-based scarcity and time-based scarcity	Reversibility of decision	Product evaluation	When consumers perceive scarcity as an indicator of value, it will enhance their evaluation of the product.
Van Herpen et al.,2014	Wine sale	×	Quantity-based scarcity	——	Product inferences, Product preference	Scarcity caused by supply increases consumer's product preference, while scarcity caused by prior demand has little impact on it.
Chen and Sun,2014	Electronic game	✓	Quantity-based scarcity	——	Purchase Intention	Product scarcity significantly enhanced purchase intention.
Wu et al.,2012	Handbag sale	×	Quantity-based scarcity	——	Purchase Intention	The impact of scarcity on purchase intention, when influenced by perceived uniqueness, is more pronounced than when scarcity is mediated by assumed expensiveness.
Gierl et al.,2008	Product retail	×	Quantity-based scarcity and time-based scarcity	The type of good	Desirability	Scarcity in quantity can help increase conspicuous consumption. If there is a significant demand for purchase, time scarcity can also be utilized. However, it is not advisable to use scarcity information signaling high demand as the source of limitation.
Jung and Kellaris,2004	Wine sale	×	Quantity-based scarcity	Context culture, product familiarity, uncertainty avoidance, and need for cognitive closure	Purchase Intention	Scarcity positively influences purchase intention, and this effect is more pronounced in countries with a lower context culture.
Lynn,1989	Art prints, wine sale	×	Quantity-based scarcity	——	Desirability	Scarcity enhances the desirability of experiences and objects
This study	**NFT**	✓	**Attribute-based scarcity**	**Consistency of rarity evaluation, impact of influencers**	**Sales and transfers**	**The digital scarcity of NFTs has a negative impact on users' trading behaviors, while positive impact on social behaviors. Besides, consistency of rarity evaluation and impact of influencers are important moderators.**

Appendix B. Results of control function approach

Variables	*logTrading*		*logSocial*	
	Model 1	Model 2	Model 3	Model 4
Scarcity* log*Influencer*		0.102***		0.162***
		(0.002)		(0.002)
Scarcity* logEvaluation		-0.026***		0.014***

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Variables		*logTrading*		*logSocial*	
		Model 1	Model 2	Model 3	Model 4
			(0.001)		(0.001)
Scarcity		-0.767***	-0.596***	0.155***	0.153***
		(0.014)	(0.014)	(0.013)	(0.014)
log*Influencer*		0.010***	0.009***	-0.027***	-0.024***
		(0.001)	(0.001)	(0.001)	(0.001)
logEvaluation		-0.001***	0.010***	0.002***	-0.003***
		(0.000)	(0.000)	(0.000)	(0.000)
logDuration		1.054***	1.058***	1.014***	1.005***
		(0.003)	(0.003)	(0.003)	(0.003)
logPrice		0.105***	0.103***	0.040***	0.041***
		(0.001)	(0.001)	(0.001)	(0.001)
logTraits	0.090***	0.064***	0.042***	-0.039***	-0.043***
	(0.000)	(0.002)	(0.002)	(0.002)	(0.002)
logRanking		-0.021***	-0.021***	-0.006***	-0.001
		(0.001)	(0.001)	(0.001)	(0.001)
Residual		0.663***	0.490***	-0.155***	-0.137***
		(0.014)	(0.015)	(0.013)	(0.014)
logNumber	-0.126***				
	(0.001)				
Constant		-6.313***	-6.321***	-6.143***	-6.128***
		(0.019)	(0.019)	(0.018)	(0.018)
Observations		849,129	849,129	849,129	849,129
R-squared		0.164	0.167	0.145	0.150

Notes: Standard errors in parentheses, *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Appendix C. Results of substituting the independent variable

Variables	*logTrading*			*logSocial*		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Scarcity logInfluencer*			0.008***			0.002***
			(0.001)			(0.000)
Scarcity logEvaluation*			0.000			-0.003***
			(0.000)			(0.000)
Scarcity		-0.024***	-0.021***		0.005***	0.003***
		(0.001)	(0.001)		(0.001)	(0.001)
logInfluencer	0.006***	0.005***	0.005***	-0.025***	-0.025***	-0.025***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
logEvaluation	0.002***	0.002***	0.002***	0.007***	0.007***	0.006***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
logDuration	1.021***	1.016***	1.016***	0.986***	0.987***	0.990***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
logPrice	0.106***	0.107***	0.106***	0.028***	0.028***	0.028***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
logTraits	-0.052***	-0.063***	-0.067***	-0.151***	-0.149***	-0.147***
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
logRanking	-0.021***	-0.020***	-0.020***	0.004***	0.004***	0.004***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Constant	-6.122***	-6.267***	-6.232***	-5.683***	-5.654***	-5.691***
	(0.019)	(0.020)	(0.020)	(0.019)	(0.019)	(0.019)
N	794,078	794,078	794,078	794,078	794,078	794,078
R^2	0.153	0.155	0.156	0.153	0.153	0.154

Notes: Standard errors in parentheses, *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Appendix D. Results of adding control variables

Variables	*logTrading*			*logSocial*		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Scarcity logInfluencer*			0.073***			0.130***
			(0.002)			(0.002)
Scarcity logEvaluation*			-0.035***			0.011***
			(0.001)			(0.001)
Scarcity		-0.123***	-0.120***		0.012***	0.023***
		(0.003)	(0.003)		(0.003)	(0.003)
logInfluencer	0.049***	0.051***	0.045***	0.002	-0.001	0.007***
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
logEvaluation	0.007***	0.006***	0.020***	0.004***	0.004***	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
logDuration	1.081***	1.086***	1.092***	1.001***	0.997***	0.989***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)

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Variables	*logTrading*			*logSocial*		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
logPrice	0.133***	0.136***	0.133***	0.052***	0.051***	0.052***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
logTraits	-0.031***	-0.023***	-0.029***	-0.085***	-0.074***	-0.077***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
logRanking	-0.010***	-0.011***	-0.012***	0.007***	0.007***	0.011***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Type	0.152***	0.154***	0.155***	0.250***	0.242***	0.235***
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
logPtraits	-0.001**	-0.000	-0.001**	-0.002***	-0.002***	-0.003***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
logSave	-0.067***	-0.067***	-0.066***	-0.041***	-0.040***	-0.039***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
logQualityholders	0.004***	-0.002	0.001	0.002	0.005***	-0.000
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Constant	-6.777***	-6.758***	-6.716***	-6.183***	-6.194***	-6.160***
	(0.019)	(0.019)	(0.019)	(0.018)	(0.018)	(0.018)
N	849,129	849,129	849,129	849,129	849,129	849,129
R^2	0.189	0.191	0.194	0.177	0.174	0.177

Notes: Standard errors in parentheses, *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

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